

jcm-strat Phase 7: how long a time step?

The time-step sweep on the Phase 6 configuration (Polvani-Kushner stratosphere, ERA5-nudged troposphere, four passive tracers, T63L95). Ten one-year runs of 2005, all judged against ERA5. Record as of 3 September 2026, branch phase7-timestep.

1. The question

Every run so far used a 12 minute step. The semi-Lagrangian dycore is unconditionally stable for advection, so the step could in principle be much longer; the handover suggested 1 to 2 hours might be reachable. Duncan's rule: double the step until the polar-vortex climatology or the tracers degrade. That step is the answer, and simulated years per day at that step is the number to report.

2. What was varied

- **The time step:** 12 (Phase 6), 30, 45, 60, 90 and 120 minutes. Nothing else changes with the step: the 6-hourly ERA5 nudging target is interpolated to whatever step is used, and all relaxation and damping rates are per unit time.
- **Departure-point iterations:** 1 (the default) or 2. The semi-Lagrangian scheme traces each grid cell backwards along the wind to find where its air came from; that trajectory is solved by a fixed-point iteration whose accuracy condition is the step times the wind shear. Two iterations were tried where one failed.

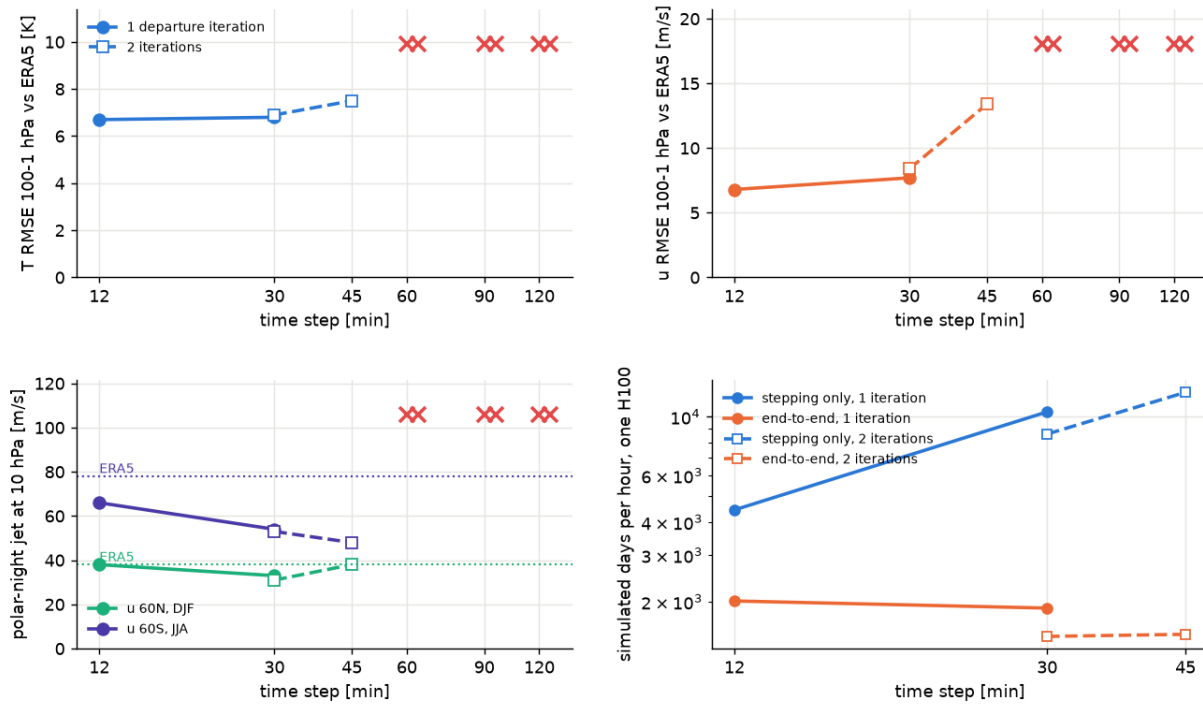
3. What happened to each run

step	iterations	outcome	T RMSE [K]	u RMSE [m/s]	jet 60N DJF	jet 60S JJA	days/hr
12 min	1	stable (Phase 6 reference)	6.7	6.8	38	66	4 445
30 min	1	stable	6.8	7.7	33	54	10 432
30 min	2	stable, same as with 1	6.9	8.4	31	53	8 602
45 min	2	stable, winds degraded	7.5	13.4	38	48	12 369
60 min	1	blew up in month 2 (T fell to 42 K); stopped by the health check	-	-	-	-	-
60 min	2	no NaN, but from May 500 m/s easterlies fill the stratosphere	24.6	178	-216	-185	-
90 min	1	blew up in month 2	-	-	-	-	-
90 min	2	no NaN, but 500 m/s easterlies from March; the clock tracer reaches 0.4 yr in a year	35.7	286	-311	-323	-
120 min	1 and 2	NaN in the first month	-	-	-	-	-
ERA5					38	78	

RMSE over 100 to 1 hPa, annual mean 2005, against ERA5. Jets: zonal wind at 10 hPa. days/hr: simulated days per wall-clock hour on one H100, model stepping only.

4. The summary figure

Phase 7: time-step sweep on the Phase 6 configuration, 2005 (ERA5 as the yardstick)



Run status (red crosses in the panels mark the runs that produced no usable year)

dt 12 min, 1 iteration:	stable — T RMSE 6.7 K, u RMSE 6.8 m/s, jets 38 / 66 m/s, 4445 days/hr
dt 30 min, 1 iteration:	stable — T RMSE 6.8 K, u RMSE 7.7 m/s, jets 33 / 54 m/s, 10432 days/hr
dt 30 min, 2 iterations:	stable — T RMSE 6.9 K, u RMSE 8.4 m/s, jets 31 / 53 m/s, 8602 days/hr
dt 45 min, 2 iterations:	stable — T RMSE 7.5 K, u RMSE 13.4 m/s, jets 38 / 48 m/s, 12369 days/hr
dt 60 min, 1 iteration:	blew up in chunk 2 (T min 42 K); stopped by the health check
dt 60 min, 2 iterations:	finite but unphysical: 500 m/s easterlies fill the stratosphere from May on
dt 90 min, 1 iteration:	blew up in chunk 2; stopped by the health check
dt 90 min, 2 iterations:	finite but unphysical: 500 m/s easterlies from March on, clock reaches only 0.4 yr
dt 120 min, 1 iteration:	NaN in chunk 1
dt 120 min, 2 iterations:	NaN in chunk 1

Figure 1. Everything on one page. Top: temperature and wind error against ERA5 as a function of the step; filled circles one departure iteration, open squares two. Bottom left: the two polar-night jets, with ERA5 dotted. Bottom right: throughput, stepping only (blue) and end-to-end including compile and output (orange). Red crosses mark the steps at which the model produced no usable year; the status lines underneath say why.

5. What the stable runs look like

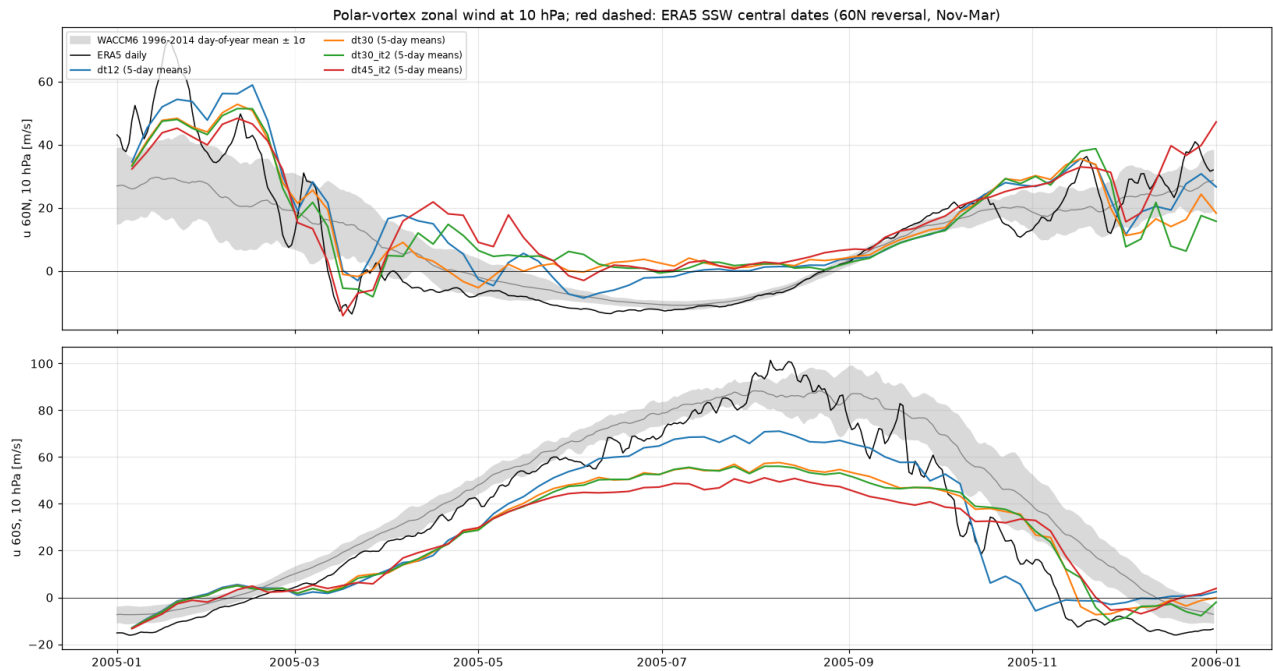


Figure 2. The 10 hPa vortex winds through 2005 for the stable runs (12, 30, 30 with two iterations, 45 minutes) against ERA5 (black) and the WACCM6 envelope (grey). The Arctic winter is reproduced at every step; the Antarctic jet loses about 12 m/s at 30 minutes and 18 m/s at 45. The two 30-minute runs lie on top of each other.

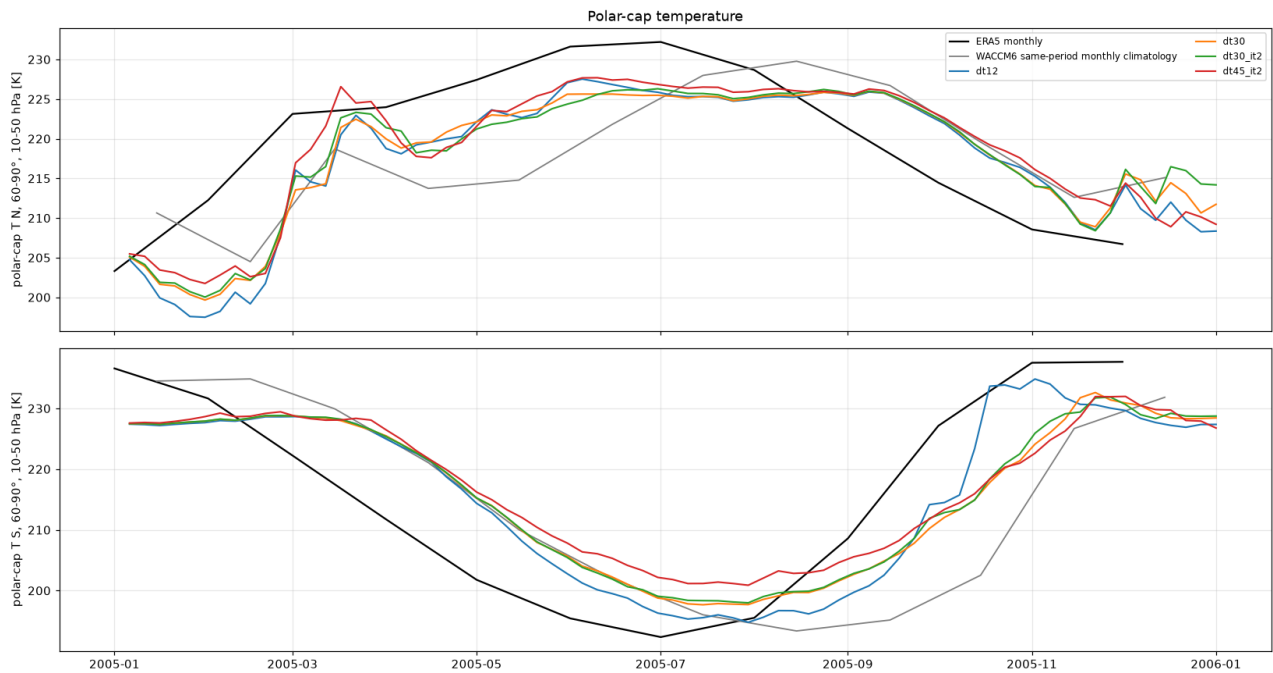


Figure 3. Polar-cap temperatures, 60 to 90 degrees, 10 to 50 hPa. The step makes almost no difference to temperature; the degradation is in the winds.

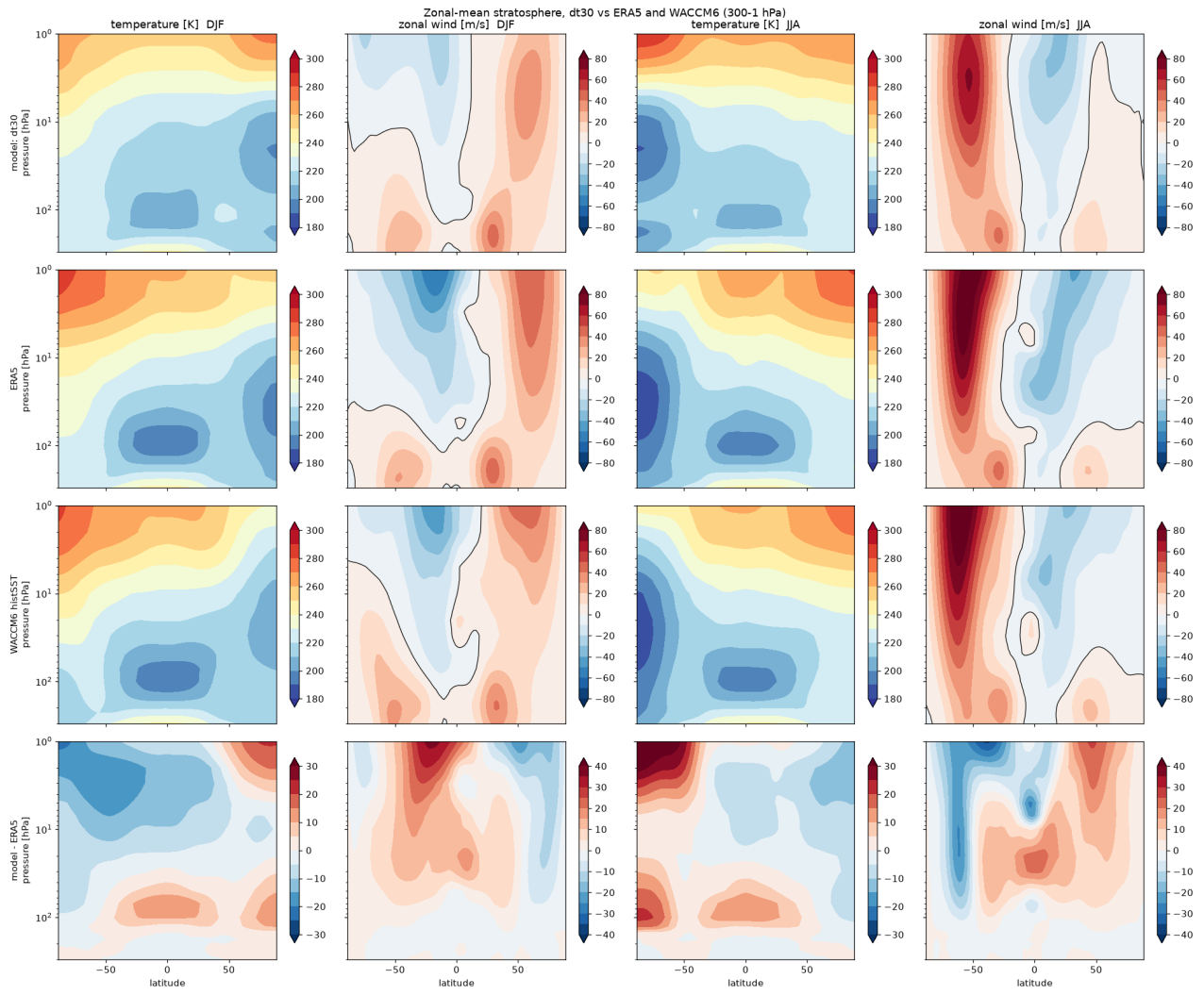


Figure 4. The 30-minute year in full: zonal-mean temperature and wind for DJF and JJA, against ERA5 and WACCM6, and the difference to ERA5 in the bottom row. Compared with the 12-minute run (Phase 6 PDF, Figure 7) the only visible change is the weaker Antarctic jet in JJA.

6. Reading and recommendation

- **The ceiling is between 45 and 60 minutes, not 2 hours.** At 60 minutes the model either blows up (one iteration) or, with two, drifts after four months into a finite but absurd state with 500 m/s easterlies. The second iteration buys protection against NaN, not accuracy: at 30 minutes it changes nothing. So the departure-point solve is not the limit; the semi-implicit coupling of the fast stratospheric jets is the more likely one, and its off-centering parameter is the next knob to sweep.
- **Cost is linear in the step.** The same 7 ms per step at 12 and 30 minutes, so 30 minutes gives 2.35 times the stepping throughput. End to end the single-year number is no better, because a new step means a fresh three-minute compile and the output volume is unchanged; over a multi-year chain those are amortised.
- **The Antarctic polar-night jet degrades first.** 66, 54, 48 m/s at 12, 30, 45 minutes while temperature and the Arctic jet hardly move. The strongest winds in the model sit exactly where the step times the shear is largest. Tracer conservation is untouched at 30 and 45 minutes.
- **Recommendation:** keep 12 minutes as the scientific default; offer 30 minutes as a fast mode for throughput and sensitivity work, stated with its 18 percent weaker Antarctic jet; do not use 45 minutes or more. Whether the jet loss at 30 minutes is acceptable for the age-of-air question is a decision for the group; a five-year chain at 30 minutes, about half an hour on one GPU, would settle it empirically.

Where things are

Runs under `runs/p7_*`, record and figures in `docs/outputs/07_timestep/`, the plumbing for the iteration count in `jcm_strat/main.py` (key `sl_departure_iterations`), the summary figure script `scripts/plot_dt_sweep.py`.